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Position detection device

Abstract

A position detection device configured to detect a position of a moving member that moves back and forth in a predetermined moving direction is provided with an exciting coil arranged to extend in the moving direction along the moving member, and a detection coil that, by an action of a magnetic field generated by the exciting coil, can detect a position, in the moving direction, of a detection object portion provided on the moving member within a predetermined detection range. A plurality of the detection coils are arranged side by side in a direction perpendicular to an extending direction of the exciting coil. A plurality of the detection object portions are provided at different positions in the moving direction so as to respectively correspond to the plurality of the detection coils, and the respective detection ranges of the plurality of the detection coils are offset in the moving direction of the moving member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is based on Japanese patent application No. 2022-136567 filed on Aug. 30, 2022, and Japanese patent application No. 2023-120025 filed on Jul. 24, 2023, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a position detection device to detect a position of a moving member that moves back and forth (i.e., reciprocating motion) in a predetermined moving direction.

BACKGROUND OF THE INVENTION

(3) Conventionally, position detection devices to detect a position of a moving member which

moves back and forth in a predetermined moving direction are used in various fields such as industrial machinery and automobiles.
(4) The electromagnetic induction linear scale (i.e., linear encoder) described in Patent Literature 1 has a coil array composed of a predetermined number of coil elements to be excited by a first AC signal, a magnetic member that relatively displaces along an axis of the coil array on the outer side of the coil array, and a detection unit that detects a position of the magnetic member relative to the coil array based on output voltages of respective coil elements. The magnetic member causes changes in the amplitude of the output voltages of the coil elements according to the positional relationship with the coil elements. The detection unit detects the position of the magnetic member relative to the coil array in absolute terms based on a phase difference between the first AC signal and a second AC signal which is obtained by synthesizing differential outputs between the coil elements.
Citation List
(5) Patent Literature 1: JP2009-2770A
SUMMARY OF THE INVENTION
(6) In the electromagnetic induction linear scale described in Patent Literature 1, the coil array must be arranged over the entire movement range of the magnetic member, and the longer the movement range of the magnetic member, the larger the size of the coil array. Therefore, it is an object of the invention to provide a position detection device that can be reduced in size.
(7) To achieve the object described above, one aspect of the invention provides a position detection device configured to detect a position of a moving member that moves back and forth in a predetermined moving direction, the position detection device comprising: an exciting coil arranged to extend in the moving direction along the moving member; and a detection coil that, by an action of a magnetic field generated by the exciting coil, can detect a position, in the moving direction, of a detection object portion provided on the moving member within a predetermined detection range, wherein a plurality of the detection coils are arranged side by side in a direction perpendicular to an extending direction of the exciting coil, wherein a plurality of the detection object portions are provided at different positions in the moving direction so as to respectively correspond to the plurality of the detection coils, and wherein the respective detection ranges of the plurality of the detection coils are offset in the moving direction of the moving member.
Advantageous Effects of the Invention
(8) According to the invention, it is possible to reduce the size of a position detection device.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a schematic diagram illustrating a vehicle on which a steer-by-wire steering system including a stroke sensor as a position detection device in an embodiment of the present invention is mounted.
(2) FIG. **2** is a cross-sectional view taken along a line A-A in FIG. **1**.
(3) FIG. **3** is a perspective view showing a rack shaft, a housing, targets and a substrate.
(4) FIG. **4**A is an overall view in which wiring patterns formed on first to fourth metal layers of the substrate are shown in a see-through manner.
(5) FIG. **4**B is a partial enlarged view of FIG. **4**A.
(6) FIGS. **5**A to **5**D are plan views respectively showing the first to fourth metal layers.
(7) FIG. **6** is a graph showing an example of a relationship between supply voltage supplied from a power supply unit to an exciting coil, induced voltage induced in a first sine wave-shaped coil element of a first detection coil, and induced voltage induced in a first cosine wave-shaped coil element of the first detection coil.

(8) FIG. **7**A is an explanatory diagram illustrating a relationship between a position of a first target and peak voltage which is a peak value of the induced voltage induced in the first sine wave-shaped coil element.
(9) FIG. **7**B is an explanatory diagram illustrating a relationship between the position of the first target and peak voltage which is a peak value of the induced voltage induced in the first cosine wave-shaped coil element.
(10) FIGS. **8**A to **8**C are explanatory diagrams illustrating a positional relationship of the first and second detection coils relative to the first and second targets.
(11) FIG. **9** is a schematic diagram illustrating a dimensional relationship between the first and second detection coils and an exciting coil on the substrate and the first and second targets.
(12) FIG. **10**A is a cross-sectional view showing the rack shaft in a modified example, together with the substrate and the housing.
(13) FIG. **10**B is a perspective view showing the rack shaft in the modified example.
(14) FIG. **11** is an explanatory diagram illustrating a circuit configuration in the second embodiment together with the rack shaft, where the wiring pattern formed on each layer of the substrate is shown in a see-through manner.
(15) FIG. **12** is an explanatory diagram illustrating a circuit configuration in the third embodiment together with the rack shaft, where the wiring pattern formed on each layer of the substrate is shown in a see-through manner.

DETAILED DESCRIPTION OF THE INVENTION

Embodiment

(16) FIG. **1** is a schematic diagram illustrating a vehicle on which a steer-by-wire steering system **10** including a stroke sensor **1** as a position detection device in an embodiment of the invention is mounted. In FIG. **1**, the steering system **10** is viewed from the rear side in a vehicle front-rear direction, the right side of the drawing corresponds to the right side in a vehicle width direction, and the left side of the drawing corresponds to the left side in the vehicle width direction. The terms “right” and “left” are sometimes used in the following description with reference to the drawings, but this expression is used for convenience of explanation and does not limit the direction of arrangement when the stroke sensor **1** is actually in use.
(17) As shown in FIG. **1**, the steering system **10** includes the stroke sensor **1**, tie rods **12** connected to steered wheels **11** (left and right front wheels), a rack shaft **13** connected to the tie rods **12**, a cylindrical housing **14** that houses the rack shaft **13**, a worm speed reduction mechanism **15** having a pinion gear **151** meshed with rack teeth **131** of the rack shaft **13**, an electric motor **16** that applies a moving force in a vehicle width direction to the rack shaft **13** through the worm speed reduction mechanism **15**, a steering wheel **17** operated by a driver, a steering angle sensor **18** to detect a steering angle of the steering wheel **17**, and a steering controller **19** that controls the electric motor **16** based on the steering angle detected by the steering angle sensor **18**.
(18) The rack shaft **13** is a moving member whose position relative to the housing **14** is detected by the stroke sensor **1**. A moving direction of the rack shaft **13** is an axial direction parallel to a central axis of the rack shaft **13**.
(19) In FIG. **1**, the housing **14** is indicated by a phantom line. The rack shaft **13** is made of, e.g., steel such as carbon steel and is supported by a pair of rack bushings **143** attached to both ends of the housing **14**. The worm speed reduction mechanism **15** has a worm wheel **152** and a worm gear **153**, and the pinion gear **151** is fixed to the worm wheel **152**. The worm gear **153** is fixed to a motor shaft **161** of the electric motor **16**.
(20) The electric motor **16** generates torque by a motor current supplied from the steering controller **19** and rotates the worm wheel **152** and the pinion gear **151** through the worm gear **153**. When the pinion gear **151** rotates, the rack shaft **13** linearly moves back and forth along the vehicle width direction, and the left and right steered wheels **11** are steered. The rack shaft **13** can move to the right and left in the vehicle width direction within a predetermined range from a neutral position at

which the steering angle is zero.
(21) In FIG. **1**, a double-headed arrow indicates a stroke range R that corresponds to the maximum travel distance of the rack shaft **13** when the steering wheel **17** is operated from one of the left and right maximum steering angles to the other maximum steering angle. The stroke sensor **1** can detect the absolute position of the rack shaft **13** relative to the housing **14** over the entire stroke range R.
(22) Configuration of Stroke Sensor **1**
(23) The stroke sensor **1** includes a target **2** which is made of a conductive metal and is attached to the rack shaft **13**, a substrate **3** arranged so as to face the target **2**, a power supply unit **4**, and a calculation unit **5**. The substrate **3** is fixed to the housing **14** so as to be parallel to the rack shaft **13**. The stroke sensor **1** detects the position of the rack shaft **13** in the axial direction (the moving direction) relative to the housing **14** based on the position of the target **2** and outputs information of the detected position to the steering controller **19**. The steering controller **19** controls the electric motor **16** so that the position of the rack shaft **13** detected by the stroke sensor **1** corresponds to the steering angle of the steering wheel **17** detected by the steering angle sensor **18**.
(24) FIG. **2** is a cross-sectional view taken along a line A-A in FIG. **1**. FIG. **3** is a perspective view showing the rack shaft **13**, a main body **141** of the housing **14**, the targets **2** and the substrate **3**.
(25) The rack shaft **13** is a rod-shaped body made of steel and having a circular cross-sectional shape. The housing **14** has the main body **141** made of metal and a lid **142** made of resin, and the lid **142** is fixed to the main body **141** by, e.g., adhesion. The main body **141**, on which a housing space **140** to house the rack shaft **13** is formed, has a U-shape in cross section, and the housing space **140** is open upward in the vertical direction. A diameter D of the rack shaft **13** is, e.g., 25 mm.
(26) A gap of, e.g., not less than 1 mm is formed between an outer peripheral surface **13***a* of the rack shaft **13** and an inner surface **140***a* of the housing space **140**. The lid **142** is formed in a flat plate shape and covers the housing space **140** from above in the vertical direction. The main body **141** is made of, e.g., a die-cast aluminum alloy. The material of the lid **142** is not necessarily limited to resin but is desirably a non-conductive material.
(27) The target **2** is a detection object portion for detection of the position of the rack shaft **13**. The target **2** may be formed on the rack shaft **13** by machining the shaft material, or the target **2** may be formed as a metal member separate from the rack shaft **13** and fixed to the rack shaft **13** by welding, etc. The target **2** is made of a material with a higher magnetic permeability than that of the rack shaft **13** or a material with a higher conductivity than that of the rack shaft **13**. When a material with a higher magnetic permeability than that of the rack shaft **13** is used for the target **2**, e.g., a magnetic material such as ferrite can be used as the material. Meanwhile, when a material with a higher conductivity than that of the rack shaft **13** is used for the target **2**, e.g., an aluminum alloy or copper alloy can be suitably used as the material. In the present embodiment, since the target **2** is provided so as to protrude from the outer peripheral surface **13***a* of the rack shaft **13** toward the substrate **3**, it is possible to obtain the functions and effects described later even when a material having the same magnetic permeability as that of the rack shaft **13** or a material having the same conductivity as that of the rack shaft **13** is used as the material of the target **2**. However, to increase the accuracy of position detection, a high magnetic permeability material with a higher magnetic permeability than that of the material of the rack shaft **13** or a high conductivity material with a higher conductivity than the material of the rack shaft **13** is desirably used as the material of the target **2**.
(28) A facing surface **2***a* of the target **2** which faces the substrate **3** is formed in a flat shape parallel to the substrate **3**. The facing surface **2***a* of the target **2** faces the front surface **3***a* of the substrate **3** in a parallel manner through an air gap G. A back surface **3***b* of the substrate **3** is fixed to the lid **142** by an adhesive **300**. A width W of the air gap G is, e.g., 1 mm. A protruding height H of the target **2** from the outer peripheral surface **13***a* of the rack shaft **13** in a direction perpendicular to the facing surface **2***a* is, e.g., 5 mm.

(29) In the present embodiment, two targets **2** are provided at different positions in an axial direction along the central axis C of the rack shaft **13**. Although the rack shaft **13** has a circular cross-sectional shape in the present embodiment, the cross-sectional shape of the rack shaft **13** is not limited to a circle and may be, e.g., a D-shape partially having a straight portion, or a polygonal shape.
(30) The substrate **3** is a four-layered substrate in which plate-shaped bases **30** made of a dielectric such as FR4 (glass fiber impregnated with epoxy resin and heat-cured) are arranged between first to fourth metal layers **301** to **304**. A thickness of each base **30** is, e.g., 0.3 mm. The first to fourth metal layers **301** to **304** are made of, e.g., copper and each have a thickness of, e.g., 18 μm. The substrate **3** has a flat rectangular shape whose long side direction coincides with the moving direction of the rack shaft **13**. The substrate **3** is not limited to a rigid substrate and may be a flexible substrate.
(31) FIG. **4**A is an overall view in which wiring patterns formed on first to fourth metal layers **301** to **304** of the substrate **3** are seen through from the back surface **3***b* side. FIG. **4**B is a partial enlarged view of FIG. **4**A. FIGS. **5**A to **5**D are plan views respectively showing the first to fourth metal layers **301** to **304** as viewed from the back surface **3***b* side. The wiring patterns shown in FIGS. **4**A, **4**B and **5**A to **5**D are merely examples, and various forms of wiring patterns can be employed as long as the substrate **3** is formed so that the effects of the invention can be obtained.
(32) In FIGS. **4**A, **4**B and **5**A to **5**D, the wiring pattern of the first metal layer **301** is indicated by solid lines, the wiring pattern of the second metal layer **302** is indicated by dashed lines, the wiring pattern of the third metal layer **303** is indicated by dashed-dotted lines, and the wiring pattern of the fourth metal layer **304** is indicated by dashed-double-dotted lines.
(33) A connector portion **35**, which has first to tenth through-holes **350** to **359** into which connector pins of a connector **6** indicated by a dashed-double-dotted line in FIG. **4**B are respectively inserted, is provided at one longitudinal end portion of the substrate **3**. The first to tenth through-holes **350** to **359** are aligned in a straight line along a lateral direction of the substrate **3**. A connector **71** (see FIG. **1**) of a cable **7** for connection to the power supply unit **4** and the calculation unit **5** is connected to the connector **6**. First to sixth vias **360** to **365** for inter-layer connection of the wiring patterns are also formed on the substrate **3**.
(34) First and second detection coils **31** and **32** to detect the positions of the targets **2** are formed on the substrate **3**. An exciting coil **33** arranged to extend in the axial direction of the rack shaft **13** along the rack shaft **13** is also formed on the substrate **3** so as to surround the first and second detection coils **31** and **32**. A sinusoidal alternating current is supplied from the power supply unit **4** to the exciting coil **33**. By an action of a magnetic field generated by the exciting coil **33**, the first and second detection coils **31** and **32** can detect the positions of the targets **2** in the moving direction of the rack shaft **13** within a predetermined detection range. The first and second detection coils **31** and **32** are arranged side by side in a direction perpendicular to an extending direction of the exciting coil **33** (perpendicular to the axial direction of the rack shaft **13**).
(35) First and second curved portions **301***a*, **301***b*, a first connector connection portion **301***c* connecting one end of the first curved portion **301***a* to the second through-hole **351**, a second connector connection portion **301***d* connecting one end of the second curved portion **301***b* to the sixth through-hole **355**, a first short-circuit line portion **301***e* connecting the first via **360** to the third via **362**, and a second short-circuit line portion **301***f* connecting the fourth via **363** to the sixth via **365** are formed on the first metal layer **301**.
(36) First and second curved portions **302***a*, **302***b*, a first connector connection portion **302***c* connecting one end of the first curved portion **302***a* to the fourth through-hole **353**, and a second connector connection portion **302***d* connecting one end of the second curved portion **302***b* to the eighth through-hole **357** are formed on the second metal layer **302**.
(37) First and second curved portions **303***a*, **303***b*, a first connector connection portion **303***c* connecting one end of the first curved portion **303***a* to the third through-hole **352**, and a second

connector connection portion **303**_d_ connecting one end of the second curved portion **303**_b_ to the seventh through-hole **356** are formed on the third metal layer **303**.

(38) First and second curved portions **304**_a_, **304**_b_, a first connector connection portion **304**_c_ connecting one end of the first curved portion **304**_a_ to the fifth through-hole **354**, and a second connector connection portion **304**_d_ connecting one end of the second curved portion **304**_b_ to the ninth through-hole **358** are formed on the fourth metal layer **304**.

(39) The other end of the first curved portion **301**_a_ of the first metal layer **301** and the other end of the first curved portion **303**_a_ of the third metal layer **303** are connected to each other by the second via **361**. Likewise, the other end of the second curved portion **301**_b_ of the first metal layer **301** and the other end of the second curved portion **303**_b_ of the third metal layer **303** are connected to each other by the fifth via **364**.

(40) The other end of the first curved portion **302**_a_ of the second metal layer **302** and the other end of the first curved portion **304**_a_ of the fourth metal layer **304** are connected to each other by the first via **360**, the third via **362** and the first short-circuit line portion **301**_e_ of the first metal layer **301**. The other end of the second curved portion **302**_b_ of the second metal layer **302** and the other end of the second curved portion **304**_b_ of the fourth metal layer **304** are connected to each other by the fourth via **363**, the sixth via **365** and the second short-circuit line portion **301**_f_ of the first metal layer **301**.

(41) The first and second curved portions **301**_a_, **302**_b_ of the first metal layer **301**, the first and second curved portions **302**_a_, **302**_b_ of the second metal layer **302**, the first and second curved portions **303**_a_, **303**_b_ of the third metal layer **303** and the first and second curved portions **304**_a_, **304**_b_ of the fourth metal layer **304** are sinusoidally curved.

(42) The first curved portion **301**_a_ of the first metal layer **301** and the first curved portion **303**_a_ of the third metal layer **303**, and also the first curved portion **302**_a_ of the second metal layer **302** and the first curved portion **304**_a_ of the fourth metal layer **304**, are symmetrical in the lateral direction of the substrate **3** across a first symmetry axis A**1** shown in FIG. **4**A. Likewise, the second curved portion **301**_b_ of the first metal layer **301** and the second curved portion **303**_b_ of the third metal layer **303**, and also the second curved portion **302**_b_ of the second metal layer **302** and the second curved portion **304**_b_ of the fourth metal layer **304**, are symmetrical in the lateral direction of the substrate **3** across a second symmetry axis A**2** shown in FIG. **4**A. The first symmetry axis A**1** and the second symmetry axis A**2** are parallel to each other and are also parallel to the axial direction of the rack shaft **13**.

(43) The first detection coil **31** has a first sine wave-shaped coil element **311** composed of the first curved portion **301**_a_ of the first metal layer **301** and the first curved portion **303**_a_ of the third metal layer **303**, and a first cosine wave-shaped coil element **312** composed of the first curved portion **302**_a_ of the second metal layer **302**, the first curved portion **304**_a_ of the fourth metal layer **304**, and the first short-circuit line portion **301**_e_ of the first metal layer **301**. That is, each of the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** has a shape of two combined sinusoidal-shaped conductor wires (the first curved portion **301**_a_ and the first curved portion **303**_a_; the first curved portion **302**_a_ and the first curved portion **304**_a_) that are symmetrical across the first symmetry axis A**1** when viewed in a direction perpendicular to the axial direction of the rack shaft **13**.

(44) The second detection coil **32** has a second sine wave-shaped coil element **321** composed of the second curved portion **301**_b_ of the first metal layer **301** and the second curved portion **303**_b_ of the third metal layer **303**, and a second cosine wave-shaped coil element **322** composed of the second curved portion **302**_b_ of the second metal layer **302**, the second curved portion **304**_b_ of the fourth metal layer **304**, and the second short-circuit line portion **301**_f_ of the first metal layer **301**. That is, each of the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** has a shape of two combined sinusoidal-shaped conductor wires (the second curved portion **301**_b_ and the second curved portion **303**_b_; the second curved portion **302**_b_ and the second

curved portion **304***b*) that are symmetrical across the second symmetry axis A**2** when viewed in the direction perpendicular to the axial direction of the rack shaft **13**.
(45) The exciting coil **33** has a rectangular shape with a pair of long-side portions **331**, **332** extending in the axial direction of the rack shaft **13** and a pair of short-side portions **333**, **334** between the pair of long-side portions **331**, **332**. In the present embodiment, the long-side portions **331**, **332** and the short-side portions **333**, **334** are formed as wiring patterns on the first metal layer **301**.
(46) Of the pair of short-side portions **333** and **334**, the short-side portion **333** on the connector portion **35** side is composed of two straight portions **333***a* and **333***b* sandwiching the first and second connector connection portions **301***c* and **301***d* of the first metal layer **301**, the first and second connector connection portions **302***c* and **302***d* of the second metal layer **302**, the first and second connector connection portions **303***c* and **303***d* of the third metal layer **303** and the first and second connector connection portions **304***c* and **304***d* of the fourth metal layer **304**, and respective ends of the two straight portions **333***a* and **333***b* are connected to the first through-hole **350** and the tenth through-hole **359** by connector connection portions **301***g* and **301***h* formed on the first metal layer **301**, as shown in FIG. **4**B.
(47) The layer on which the exciting coil **33** is formed is not limited to the first metal layer **301** and may be any of the second to fourth metal layers **302** to **304**, or the exciting coil **33** may be formed across plural metal layers. Alternatively, the exciting coil may be formed separately from the substrate **3**. Although the exciting coil **33** makes one turn around the first and second detection coils **31** and **32** in the present embodiment, the exciting coil may be formed to make plural turns around the first and second detection coils **31** and **32**.
(48) Of the two targets **2** provided on the rack shaft **13**, one target **2** is provided so as to correspond to the first detection coil **31**, and the other target **2** is provided so as to correspond to the second detection coil **32**. Hereinafter, the one target **2** corresponding to the first detection coil **31** is referred to as a first target **21**, and the other target **2** corresponding to the second detection coil **32** is referred to as a second target **22**.
(49) When a direction perpendicular to the front surface **3***a* and the back surface **3***b* of the substrate **3** is defined as the substrate normal direction, the first target **21** is provided at a position lining up with the first detection coil **31** in the substrate normal direction and not lining up with the second detection coil **32** in the substrate normal direction. Likewise, the second target **22** is provided at a position lining up with the second detection coil **32** in the substrate normal direction and not lining up with the first detection coil **31** in the substrate normal direction.
(50) Induced voltage is generated in the first and second detection coils **31** and **32** due to linkage with the magnetic flux of the magnetic field generated by the exciting coil **33**. When the target **2** is made of a material with a higher magnetic permeability than that of the rack shaft **13**, the magnetic flux intensively flows through the target **2** and the substrate **3** has a higher magnetic flux density at a portion facing the target **2** than at other portions. Meanwhile, when the target **2** is made of a material with a higher conductivity than that of the rack shaft **13**, the density of the magnetic flux linking with the first and second detection coils **31** and **32** becomes low due to an eddy current generated in the target **2** by an AC magnetic field, and the substrate **3** has a lower magnetic flux density at the portion facing the target **2** than at other portions. Therefore, the magnitude of voltage induced in the first and second detection coils **31** and **32** changes according to the position of the target **2** relative to the substrate **3**. In this regard, when a material with a higher magnetic permeability than that of the rack shaft **13** is used as the material of the target **2**, it is desirable to use a magnetic material which has high electric resistance and in which an eddy current is not easily generated.
(51) A voltage having the same cycle as that of an alternating current supplied from the power supply unit **4** to the exciting coil **33** is induced in the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** of the first detection coil **31**, and a peak value of this

induced voltage changes according to the position of the first target **21** relative to the substrate **3**. Likewise, a voltage having the same cycle as that of the alternating current supplied from the power supply unit **4** to the exciting coil **33** is induced in the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** of the second detection coil **32**, and a peak value of this induced voltage changes according to the position of the second target **21** relative to the substrate **3**. The peak value of voltage here means the absolute maximum value of the voltage in one cycle of the alternating current supplied to the exciting coil **33**.
(52) Phases of the peak values of the voltages induced in the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** of the first detection coil **31** during when the rack shaft **13** moves from one moving end to another moving end in the axial direction are different from each other. Likewise, phases of the peak values of the voltages induced in the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** of the second detection coil **32** during when the rack shaft **13** moves from the one moving end to the other moving end in the axial direction are also different from each other.
(53) In the present embodiment, the phase difference between the peak values of the voltages induced in the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** of the first detection coil **31** and the phase difference between the peak values of the voltages induced in the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** of the second detection coil **32** are each 90°. The voltages induced in the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** of the first detection coil **31** and the voltages induced in second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** of the second detection coil **32** are each output as output voltage to the calculation unit **5** through the cable **7**.
(54) As shown in FIG. **4**A, first and second buffer regions E**1** and E**2** to suppress voltages induced in the first sine wave-shaped coil element **311**, the first cosine wave-shaped coil element **312**, the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** by a magnetic flux generated by a current flowing through the pair of short-side portions **333**, **334** are provided between the pair of short-side portions **333**, **334** of the exciting coil **33** and the first detection coil **31**/the second detection coil **32**.
(55) In the first buffer region E**1**, to suppress induced voltage generated when the first target **21** or the second target **22** is present at a position aligned with the first buffer region E**1** in the substrate normal direction, the first connector connection portions **301***c*, **302***c*, **303***c*, **304***c* respectively formed on the first to fourth metal layers **301** to **304** overlap in a thickness direction of the substrate **3** and the second connector connection portions **301***d*, **302***d*, **303***d*, **304***d* respectively formed on the first to fourth metal layers **301** to **304** overlap in the thickness direction of the substrate **3**.
(56) FIG. **6** is a graph showing an example of a relationship between supply voltage V**0** supplied from the power supply unit **4** to the exciting coil **33**, induced voltage VS**1** induced in the first sine wave-shaped coil element **311** of the first detection coil **31** and induced voltage VC**1** induced in the first cosine wave-shaped coil element **312**. In the graph of FIG. **6**, the horizontal axis is the time axis, and the left and right vertical axes indicate the supply voltage V**0** and the induced voltages VS**1**, VC**1**. A high-frequency AC voltage of, e.g., about 1 MHz is supplied as the supply voltage V**0** to the exciting coil **33**.
(57) The supply voltage V**0** and the induced voltages VS**1**, VC**1** are in phase with each other in the example shown in FIG. **6**. However, the induced voltage VS**1** induced in the first sine wave-shaped coil element **311** switches between in-phase and antiphase at the time that the first target **21** passes through a position corresponding to an intersection point between the first curved portion **301***a* of the first metal layer **301** and the first curved portion **303***a* of the third metal layer **303** as viewed in the substrate normal direction. Likewise, the induced voltage VC**1** induced in the first cosine wave-shaped coil element **312** switches between in-phase and antiphase at the time that the first target **21**

passes through a position corresponding to an intersection point between the first curved portion **302***a* of the second metal layer **302** and the first curved portion **304***a* of the fourth metal layer **304** as viewed in the substrate normal direction.
(58) FIG. **7**A is an explanatory diagram schematically illustrating a relationship between the position of the first target **21** and peak voltage VSP**1** which is the peak value of the induced voltage VS**1** induced in the first sine wave-shaped coil element **311**. FIG. **7**B is an explanatory diagram schematically illustrating a relationship between the position of the first target **21** and peak voltage VCP**1** which is the peak value of the induced voltage VC**1** induced in the first cosine wave-shaped coil element **312**.
(59) In the graphs of the peak voltages VSP**1** and VCP**1** shown in FIGS. **7**A and **7**B, the horizontal axis indicates the position of the center of the first target **21**. P**1** on the horizontal axis indicates the position of the central portion of the first target **21** when the left end of the first target **21** coincides with the left ends of the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312**. P**2** on the horizontal axis indicates the position of the central portion of the first target **21** when the right end of the first target **21** coincides with the right ends of the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312**. In FIGS. **7**A and **7**B, the first target **21** located at the position P**1** is indicated by a dashed-dotted line, and the first target **21** located at the position P**2** is indicated by a dashed-double-dotted line.
(60) In addition, in FIGS. **7**A and **7**B, the magnetic field generated by an eddy current in the first target **21** is schematically indicated by arrows. Due to a difference in magnetic field strength between a portion facing the first target **21** in the substrate normal direction and a portion not facing the first target **21**, the output voltages of the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** change according to the position of the rack shaft **13**.
(61) In the graph shown in FIG. **7**A, the peak voltage VSP**1** has a positive value when the induced voltage VS**1** induced in the first sine wave-shaped coil element **311** is in phase with the supply voltage V**0** supplied to the exciting coil **33**, and has a negative value when in antiphase. Likewise, in the graph shown in FIG. **7**B, the peak voltage VCP**1** has a positive value when the induced voltage VC**1** induced in the first cosine wave-shaped coil element **312** is in phase with the supply voltage V**0** supplied to the exciting coil **33**, and has a negative value when in antiphase.
(62) When the rack shaft **13** moves in one direction, the peak voltage VSP**1** changes sinusoidally and the peak voltage VCP**1** changes cosinusoidally as shown in FIGS. **7**A and **7**B during when the entire portion of the first target **21** in the axial direction of the rack shaft **13** is overlapping the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** in the substrate normal direction. Thus, the calculation unit **5** can determine the position of the rack shaft **13** by calculation based on the output voltages of the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312**.
(63) Likewise for the second detection coil **32**, when the rack shaft **13** moves in one direction, a peak voltage VSP**2**, which is a peak value of an induced voltage VS**2** induced in the second sine wave-shaped coil element **321**, changes sinusoidally and a peak voltage VCP**2**, which is a peak value of an induced voltage VC**2** induced in the second cosine wave-shaped coil element **322**, changes cosinusoidally during when the entire portion of the second target **22** in the axial direction of the rack shaft **13** is overlapping the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** in the substrate normal direction. Thus, the calculation unit **5** can determine the position of the rack shaft **13** by calculation based on the output voltages of the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322**. The details of how the calculation unit **5** calculates the position of the rack shaft **13** will be described later.
(64) FIGS. **8**A to **8**C are explanatory diagrams illustrating a positional relationship of the first and second detection coils **31**, **32** relative to the first and second targets **21**, **22** in the substrate normal direction. In FIGS. **8**A to **8**C, the first and second detection coils **31**, **32** and the exciting coil **33** are

seen through the base **30** from the back surface **3***b* side of substrate **3** and are superimposed on the rack shaft **13** and the first and second targets **21**, **22** in the substrate normal direction.
(65) FIGS. **8**A to **8**C also show a center point C**1** of a range on the substrate **3** where the first and second detection coils **31** and **32** are formed, center points C**21**, C**22** of the respective facing surfaces **2***a* of the first and second targets **21**, **22**, and a midpoint C**2** between the center points C**21** and C**22**.
(66) FIG. **8**A shows a state in which the rack shaft **13** is in the neutral position at which the steering angle of the steering wheel **17** is zero, and the center point C**1** coincides with the midpoint C**2**. In this state, the left end of the first target **21** coincides with the left end of the first detection coil **31**, and the right end of the second target **22** coincides with the right end of the second detection coil **32**.
(67) FIG. **8**B shows a state in which the right end of the first target **21** coincides with the right end of the first detection coil **31**. The distance in the axial direction of the rack shaft **13** between the center point C**1** and the midpoint C**2** in this state is defined as L**1**. The range of the distance L**1** is a detection range where the position of the first target **21** can be detected by the first detection coil **31**.
(68) FIG. **8**C shows a state in which the left end of the second target **22** coincides with the left end of the second detection coil **32**. The distance in the axial direction of the rack shaft **13** between the center point C**1** and the midpoint C**2** in this state is defined as L**2**. The range of the distance L**2** is a detection range where the position of the second target **22** can be detected by the second detection coil **32**.
(69) In this manner, the detection range where the position of the first target **21** can be detected by the first detection coil **31** and the detection range where the position of the second target **22** can be detected by the second detection coil **32** are offset in the moving direction of the rack shaft **13**. In addition, when the rack shaft **13** is in the neutral position, the position of the first target **21** can be detected by the first detection coil **31** and the position of the second target **22** can be detected by the second detection coil **32**. That is, the detection range where the position of the first target **21** can be detected by the first detection coil **31** and the detection range where the position of the second target **22** can be detected by the second detection coil **32** partially overlap and are continuous in the moving direction of the rack shaft **13**.
(70) The calculation unit **5** can calculate and determine the absolute position of the rack shaft **13** in an axial range of length L**1**+L**2**, with the neutral position of the rack shaft **13** at the center. The length L**1**+L**2** is the same as or longer than the length of the stroke range R of the rack shaft **13** (see FIG. **1**), and the stroke sensor **1** can detect the absolute position of the rack shaft **13** over the entire stroke range R.
(71) FIG. **9** is a schematic diagram illustrating a dimensional relationship between the first and second detection coils **31**, **32** and the exciting coil **33** on the substrate **3** and the first and second targets **21**, **22**. In FIG. **9**, the axial direction of the rack shaft **13** is the x-axis direction, and a direction parallel to the substrate **3** and perpendicular to the x-axis direction is the y-axis direction. In addition, in FIG. **9**, the dimensions of the first and second detection coils **31** and **32** and the exciting coil **33** in the y-axis direction are enlarged two times, and a spacing between the first detection coil **31** and the second detection coil **32** is enlarged in the y-axis direction.
(72) The origin of the x-axis is a position at which the center point C**21** of the facing surface **2***a* of the first target **21** is located when in the state shown in FIG. **8**C, and the origin of the y-axis is a position at which the long-side portion **332** of the exciting coil **33** is located. A coordinate point X**1** on the x-axis indicates a position in the x-axis direction at which the center point C**21** of the facing surface **2***a* of the first target **21** is located when in the state shown in FIG. **8**A, and a coordinate point X**2** on the x-axis indicates a position in the x-axis direction at which the center point C**22** of the facing surface **2***a* of the second target **22** is located when in the state shown in FIG. **8**A. A coordinate point X**3** on the x-axis indicates a position in the x-axis direction at which the center

point C**22** of the facing surface **2***a* of the second target **22** is located when in the state shown in FIG. **8**B. A coordinate point C**2**′ is a position at which the midpoint C**2** between the center points C**21** and C**22** of the first and second targets **21** and **22** is located when in the state shown in FIG. **8**C, and a coordinate point C**2**″ is a position at which the midpoint C**2** between the center points C**21** and C**22** of the first and second targets **21** and **22** is located when in the state shown in FIG. **8**B.
(73) In FIG. **9**, L is the length of the first and second detection coils **31** and **32** in the x-axis direction. u is a ratio of the length of the first and second targets **21** and **22** in the x-axis direction to L. Δp**1** is a distance in the x-axis direction from the left ends of the first and second detection coils **31** and **32** to the center point C**21** of the first target **21**. Δp**2** is a distance in the x-axis direction from the left ends of the first and second detection coils **31** and **32** to the center point C**22** of the second target **22**.
(74) When the first target **21** overlaps the first sine wave-shaped coil element **311** and the first cosine wave-shaped coil element **312** in the substrate normal direction, Δp**1** can be obtained by the following equation (1).
(75) $$p_1 = L\frac{\tan^{-1}\left(\frac{V_{SP1}}{V_{CP1}}\right)}{2} \quad (1)$$
(76) When the second target **22** overlaps the second sine wave-shaped coil element **321** and the second cosine wave-shaped coil element **322** in the substrate normal direction, Δp**2** can be obtained by the following equation (2).
(77) $$p_2 = L\frac{\tan^{-1}\left(\frac{V_{SP2}}{V_{CP2}}\right)}{2} \quad (2)$$
(78) The calculation unit **5** can determine the absolute position of the rack shaft **13** by calculation using the equation (1) or the equation (2). The movement distance of the rack shaft **13** in the x-axis direction over which the calculation unit **5** can determine the absolute position of the rack shaft **13** by the equation (1) and the movement distance of the rack shaft **13** in the x-axis direction over which the calculation unit **5** can determine the absolute position of the rack shaft **13** by the formula (2) are each (1−u)L. Thus, in the present embodiment, the overall movement distance of the rack shaft **13** over which the calculation unit **5** can determine the absolute position of the rack shaft **13** is 2(1−u)L. In this regard, u is a value smaller than 0.5. The smaller the value of u, the longer the distance over which the absolute position of the rack shaft **13** can be detected, but if the value of u is too small, the induced voltages become small and errors are likely to be larger. Therefore, the value of u is desirably, e.g., not less than 0.01 and less than 0.5.
Functions and Effects of the Embodiment
(79) In the embodiment described above, it is possible to detect the absolute position of the rack shaft **13** over a longer distance than the length L of the first and second detection coils **31** and **32** in the x-axis direction. In other words, the length L of the first and second detection coils **31** and **32** in the x-axis direction, which is the axial direction of the rack shaft **13**, can be shorter than the maximum travel distance of the rack shaft **13**, which allows for size reduction of the stroke sensor **1**. In addition, in the present embodiment, it is possible to reduce the size of the substrate **3** on which the first and second detection coils **31** and **32** are formed.
Modified Example
(80) Next, a modified example of the embodiment will be described in reference to FIG. **10**. FIG. **10**A is a cross-sectional view showing the rack shaft **13** in the modified example, together with the substrate **3** and the housing **14**. FIG. **10**B is a perspective view showing the rack shaft **13**, the substrate **3** and the main body **141** of the housing **14** in the modified example. The substrate **3** is configured in the same manner as in the embodiment described above and is connected to the power supply unit **4** and the calculation unit **5** through the connector **6** and the cable **7**.
(81) In the embodiment described above, the example in which the detection object portion for detection of the position of the rack shaft **13** is the target **2** protruding toward the substrate **3**

beyond the outer peripheral surface **13***a* of the rack shaft **13** has been described. In contrast, in this modified example, the detection object portion is a recessed portion **130** provided on the rack shaft **13**. The rack shaft **13** has a conductivity, and the recessed portion **130** is recessed from the outer peripheral surface **13***a* of the rack shaft **13**.
(82) Two recessed portions **130** are provided on the rack shaft **13** at different positions in the axial direction of the rack shaft **13** so as to respectively correspond to the first and second detection coils **31** and **32**, and the respective detection ranges of the first and second detection coils **31** and **32** are offset in the moving direction of the rack shaft **13**. The positions where the two recessed portions **130** are provided are the same as the positions where the first and second targets **21** and **22** are provided in the embodiment described above.
(83) Of the two recessed portions **130**, one recessed portion **130** is provided at a position lining up with the first detection coil **31** in the substrate normal direction and not lining up with the second detection coil **32** in the substrate normal direction. The other recessed portion **130** is provided at a position lining up with the second detection coil **32** in the substrate normal direction and not lining up with the first detection coil **31** in the substrate normal direction. In the substrate **3**, the magnetic field strength at a portion facing the recessed portion **130** is weaker than the magnetic field strength at a portion not facing the recessed portion **130**. The output voltages of the first and second detection coils **31** and **32** change according to the position of the rack shaft **13** due to a difference in magnetic field strength between the portion facing the recessed portion **130** and the portion not facing the recessed portion **130**.
(84) Also in this modified example, it is possible to detect the absolute position of the rack shaft **13** over a longer distance than the length of the first and second detection coils **31** and **32**, in the same manner as the embodiment described above. In addition, the distance between the rack shaft **13** and the substrate **3** can be reduced as compared to the embodiment described above, allowing for further size reduction. In this regard, to suppress a decrease in rigidity of the rack shaft **13** caused by providing the recessed portions **130**, the recessed portions **130** may be filled with, e.g., a non-conductive filler made of hard resin, etc.
Second Embodiment
(85) Next, the second embodiment of the invention will be described in reference to FIG. **11**. FIG. **11** is a configuration diagram illustrating a stroke sensor **1**A in the second embodiment. FIG. **11** shows the circuit configuration of the stroke sensor **1**A superimposed on the rack shaft **13**, where the wiring pattern formed on each layer of a substrate **3**A is shown in a see-through manner.
(86) The first and second detection coils **31**, **32**, third and fourth detection coils **37**, **38** and an exciting coil **39** are formed on the substrate **3**A. The substrate **3**A is a four-layered substrate in the same manner as the first embodiment, and the first and second detection coils **31**, **32**, the third and fourth detection coils **37**, **38**, and the exciting coil **39** are distributed between layers.
(87) The shape and size of the first and second detection coils **31** and **32** are the same as in the first embodiment. The third and fourth detection coils **37** and **38** have third and fourth sine wave-shaped coil elements **371**, **381** and third and fourth cosine wave-shaped coil elements **372**, **382** which are different in size from the first and second sine wave-shaped coil elements **311**, **321** and the first and second cosine wave-shaped coil elements **312**, **322** and have a shorter length in the axial direction of the rack shaft **13**.
(88) The exciting coil **39** is formed so as to surround the first and second detection coils **31**, **32** and the third and fourth detection coils **37**, **38** and is connected to the power supply unit **4** through the connector **6** and the cable **7**. The first and second detection coils **31**, **32** and the third and fourth detection coils **37**, **38** are connected to the calculation unit **5** through the connector **6** and the cable **7**.
(89) As the plurality of detection object portions for detection of the position of the rack shaft **13**, plural first targets **21** corresponding to the first detection coil **31**, plural second targets **22** corresponding to the second detection coil **32**, plural third targets **23** corresponding to the third

detection coil **37**, and plural fourth targets **24** corresponding to the fourth detection coil **38** are provided on the rack shaft **13** and are respectively aligned in the axial direction of the rack shaft **13**.
(90) The plural first targets **21** are spaced apart in the axial direction of the rack shaft **13** at a predetermined interval. The plural second targets **22** are spaced apart in the axial direction of the rack shaft **13** at the same predetermined interval as the plural first targets **21**. The plural third targets **23** are spaced apart in the axial direction of the rack shaft **13** at predetermined intervals narrower than the interval of the plural first targets **21**. The plural fourth targets **24** are spaced apart in the axial direction of the rack shaft **13** at the same predetermined intervals as the plural third targets **23**.
(91) During when the rack shaft **13** moves from the one moving end to the other moving end in the axial direction, the first detection coil **31** faces the plural first targets **21** in the substrate normal direction and the second detection coil **32** faces the plural second targets **22** in the substrate normal direction. In addition, during when the rack shaft **13** moves from the one moving end to the other moving end in the axial direction, the third detection coil **37** faces the plural third targets **23** in the substrate normal direction and the fourth detection coil **38** faces the plural fourth targets **24** in the substrate normal direction.
(92) The detection range where the positions of the plural first targets **21** in the axial direction of the rack shaft **13** can be detected by the first detection coil **31** and the detection range where the positions of the plural second targets **22** in the axial direction of the rack shaft **13** can be detected by the second detection coil **32** are partially overlap and are continuous in the moving direction of the rack shaft **13**. Likewise, the detection range where the positions of the plural third targets **23** in the axial direction of the rack shaft **13** can be detected by the third detection coil **37** and the detection range where the positions of the plural fourth targets **24** in the axial direction of the rack shaft **13** can be detected by the fourth detection coil **38** are partially overlap and are continuous in the moving direction of the rack shaft **13**.
(93) When the rack shaft **13** moves in one direction, the distance that the rack shaft **13** moves during one cycle of change in the peak values of the voltages induced in the first and second detection coils **31** and **32** is different from the distance that the rack shaft **13** moves during one cycle of change in the peak values of the voltages induced in the third and fourth detection coils **37** and **38**. In the second embodiment, in proportion to the difference in length in the axial direction of the rack shaft **13** between the first and second detection coils **31**, **32** and the third and fourth detection coils **37**, **38**, the distance that the rack shaft **13** moves during one cycle of change in the peak values of the voltages induced in the third and fourth detection coils **37** and **38** is shorter than the distance that the rack shaft **13** moves during one cycle of change in the peak values of the voltages induced in the first and second detection coils **31** and **32**.
(94) When the distance that the rack shaft **13** moves during one cycle of change in the peak values of the voltages induced in the first and second detection coils **31** and **32** is defined as X**12** and the distance that the rack shaft **13** moves during one cycle of change in the peak values of the voltages induced in the third and fourth detection coils **37** and **38** is defined as X**34**, X**34** is shorter than X**12** and X**12** is a non-integer multiple of X**34**. Thus, the calculation unit **5** can calculate the absolute position of the rack shaft **13** over a longer distance than the length L**1**+L**2** in the first embodiment by comparing the position of the first target **21** or the second target **22** determined based on an output signal of the first detection coil **31** or the second detection coil **32** with the position of the third target **23** or the fourth target **24** determined based on an output signal of the third detection coil **37** or the four detection coil **38**. In addition, if the length of the stroke range of the rack shaft **13** is the same as that in the first embodiment, the length of the substrate **3**A in the axial direction of the rack shaft **13** can be even shorter than the length of the substrate **3** in the axial direction in the first embodiment.
Third Embodiment
(95) Next, the third embodiment of the invention will be described in reference to FIG. **12**. FIG. **12**

is a configuration diagram illustrating a stroke sensor **1**B in the third embodiment. FIG. **12** shows the circuit configuration of the stroke sensor **1**B superimposed on the rack shaft **13**, where the wiring pattern formed on each layer of the substrate **3** is shown in a see-through manner.
(96) The substrate **3** is configured in the same manner as in the first embodiment. Plural first targets **21**, which are provided as the detection object portions corresponding to the first detection coil **31** and are spaced apart in the axial direction of the rack shaft **13**, and plural second targets **22**, which are provided as the detection object portions corresponding to the second detection coil **32** and are spaced apart in the axial direction of the rack shaft **13**, are provided on the rack shaft **13**.
(97) The detection range where the positions of the plural first targets **21** can be detected by the first detection coil **31** and the detection range where the positions of the second targets **22** can be detected by the second detection coil **32** are partially overlap and are continuous in the moving direction of the rack shaft **13**. During when the rack shaft **13** moves from the one moving end to the other moving end, the plural first targets **21** face the first detection coil **31** and the plural second targets **22** face the second detection coil **32**.
(98) Each of the plural first targets **21** has a different length in the axial direction of the rack shaft **13**. Likewise, each of the plural second targets **22** has a different length in the axial direction of the rack shaft **13**.
(99) A magnitude of the induced voltage induced in the first detection coil **31** differs according to the length of the first target **21**, and a magnitude of the induced voltage induced in the second detection coil **32** differs according to the length of the second target **22**. Thus, based on the magnitude of the induced voltage, the calculation unit **5** can determine which of the plural first targets **21** is facing the first detection coil **31** and which of the plural second targets **22** is facing the second detection coil **32**. It is thereby possible to calculate the absolute position of the rack shaft **13** over a longer distance than the length L**1**+L**2** in the first embodiment.
SUMMARY OF THE EMBODIMENTS
(100) Technical ideas understood from the embodiments will be described below citing the reference signs, etc., used for the embodiments. However, each reference sign, etc., described below is not intended to limit the constituent elements in the claims to the members, etc., specifically described in the embodiments.
(101) According to the first feature, a position detection device (the stroke sensor **1**) configured to detect a position of a moving member (the rack shaft **13**) that moves back and forth in a predetermined moving direction, the position detection device **1**, **1**A, **1**B comprising: an exciting coil **33**, **39** arranged to extend in the moving direction along the moving member **13**; and a detection coil **31**, **32**, **37**, **38** that, by an action of a magnetic field generated by the exciting coil **33**, **39**, can detect a position, in the moving direction, of a detection object portion (the target **2**, the first to fourth targets **21-24**, the recessed portion **130**) provided on the moving member **13** within a predetermined detection range, wherein a plurality of the detection coils **31**, **32**, **37**, **38** are arranged side by side in a direction perpendicular to an extending direction of the exciting coil **33**, **39**, wherein a plurality of the detection object portions **2**, **21-24**, **130** are provided at different positions in the moving direction so as to respectively correspond to the plurality of the detection coils **31**, **32**, **37**, **38**, and wherein the respective detection ranges of the plurality of the detection coils **31**, **32**, **37**, **38** are offset in the moving direction of the moving member **13**.
(102) According to the second feature, in the position detection device **1** as described by the first feature, the detection coil **31**, **32**, **37**, **38** comprises a pair of coil elements **311**, **312**, **321**, **322**, **371**, **372**, **381**, **382** with output voltages that change according to the position of the moving member **13**, and phases of the respective output voltages of the pair of coil elements **311**, **312**, **321**, **322**, **371**, **372**, **381**, **382** during movement of the moving member **13** is different from each other.
(103) According to the third feature, in the position detection device **1**, **1**A, **1**B as described by the second feature, each of the pair of coil elements **311**, **312**, **321**, **322**, **371**, **372**, **381**, **382** has a shape of two combined sinusoidal-shaped conductor wires **301***a*, **301***b*, **302***a*, **302***b*, **303***a*, **303***b*, **304***a*,

304*b* that are symmetrical across a symmetry axis A**1**, A**2** parallel to the moving direction when viewed in a direction perpendicular to the moving direction, and the output voltage changes according to the position of the moving member **13** due to a difference in magnetic field strength between a portion facing the detection object portion **2**, **21-24**, **130** and a portion not facing the detection object portion **2**, **21-24**, **130**.

(104) According to the fourth feature, in the position detection device **1**, **1**A, **1**B as described by the first feature, the exciting coil **33**, **39** and the plurality of the detection coils **31**, **32**, **37**, **38** are formed on one substrate **3**, **3**A, and the exciting coil **33**, **39** is formed on the substrate **3**, **3**A so as to surround the plurality of the detection coils **31**, **32**, **37**, **38**.

(105) According to the fifth feature, in the position detection device **1**, **1**A, **1**B as described by the first feature, the detection object portion **2**, **21-24** comprises a conductive member protruding toward the detection coil **31**, **32**, **37**, **38**.

(106) According to the sixth feature, in the position detection device **1** as described by the first feature, the moving member **13** has conductivity, and wherein the detection object portion comprises a recessed portion **130** provided on the moving member **13**.

(107) According to the seventh feature, in the position detection device **1**, **1**A, **1**B as described by any one of the first to sixth features, the plurality of the detection coils comprise a first detection coil **31** and a second detection coil **32**, and the detection range of the first detection coil **31** and the detection range of the second detection coil **32** partially overlap and are continuous in the moving direction.

(108) According to the eighth feature, in the position detection device **1**, **1**A, **1**B as described by the seventh feature, during when the moving member **13** moves from one moving end to another moving end in the moving direction, each of the first detection coil **31** and the second detection coil **32** faces a plurality of the detection object portions **2**, **21**, **22** that are provided so as to be spaced apart in the moving direction.

(109) According to the ninth feature, in the position detection device **1**B as described by the eighth feature, the plurality of the detection object portions **21**, which face the first detection coil **31** during when the moving member **13** moves from the one moving end to the other moving end, each have a different length in the moving direction, and the plurality of the detection object portions **22**, which face the second detection coil **32** during when the moving member **13** moves from the one moving end to the other moving end, each have a different length in the moving direction.

(110) According to the tenth feature, in the position detection device **1**A as described by any one of the first to sixth features, among the plurality of the detection coils **31**, **32**, **37**, **38**, at least one detection coil **31**, **32** and at least one other detection coil **37**, **38** have different lengths from each other in the moving direction, and wherein a cycle of an output signal of the at least one detection coil **31**, **32** and a cycle of an output signal of the at least one other detection coil **37**, **38** during movement of the moving member **13** are different from each other.

(111) According to the eleventh feature, in the position detection device **1**A as described by the tenth feature, during when the moving member **13** moves from the one moving end to the other moving end in the moving direction, each of the plurality of detection coils **31**, **32**, **37**, **38** faces a plurality of the detection object portions **21-24** that are provided so as to be spaced apart in the moving direction.

(112) Although the embodiments and modification of the invention have been described, the invention according to claims is not to be limited to the embodiments and modification described above. Further, please note that not all combinations of the features described in each of the embodiments and modification are necessary to solve the problem of the invention.

(113) In addition, the invention can be appropriately modified and implemented without departing from the gist thereof. For example, in the above-described second embodiment, although the example in which the first to fourth targets **21** to **24** are provided as the detection object portions on

the rack shaft **13** has been described, recessed portions provided on the rack shaft **13** as in the above-described modified example may be used as the detection object portions in place of the first to fourth targets **21** to **24**. Likewise, in the third embodiment, in place of the plural first and second targets **21** and **22** having different lengths in the axial direction of the rack shaft **13**, plural recessed portions having different lengths in the axial direction of the rack shaft **13** may be provided as the detection object portions on the rack shaft **13**.
(114) In addition, the moving member subjected to position detection by the stroke sensor **1**, **1**A, **1**B is not limited to the rack shaft **13** of the steering system **10** and may be a shaft to be mounted on an automobile or a shaft not to be mounted on an automobile. In addition, the shape of the moving member is not limited to a shaft shape and can be various shapes such as a flat plate shape.

Claims

1. A position detection device configured to detect a position of a moving member that moves back and forth in a predetermined moving direction, the position detection device comprising: an exciting coil arranged to extend in the moving direction along the moving member; and a detection coil that, by an action of a magnetic field generated by the exciting coil, can detect a position, in the moving direction, of a detection object portion provided on the moving member within a predetermined detection range, wherein a plurality of the detection coils are arranged side by side in a direction perpendicular to an extending direction of the exciting coil, wherein a plurality of the detection object portions are provided at different positions in the moving direction so as to respectively correspond to the plurality of the detection coils, and wherein the respective detection ranges of the plurality of the detection coils are offset in the moving direction of the moving member.
2. The position detection device according to claim 1, wherein the detection coil comprises a pair of coil elements with output voltages that change according to the position of the moving member, and phases of the respective output voltages of the pair of coil elements during movement of the moving member is different from each other.
3. The position detection device according to claim 2, wherein each of the pair of coil elements has a shape of two combined sinusoidal-shaped conductor wires that are symmetrical across a symmetry axis parallel to the moving direction when viewed in a direction perpendicular to the moving direction, and the output voltage changes according to the position of the moving member due to a difference in magnetic field strength between a portion facing the detection object portion and a portion not facing the detection object portion.
4. The position detection device according to claim 1, wherein the exciting coil and the plurality of the detection coils are formed on one substrate, and wherein the exciting coil is formed on the substrate so as to surround the plurality of the detection coils.
5. The position detection device according to claim 1, wherein the detection object portion comprises a conductive member protruding toward the detection coil.
6. The position detection device according to claim 1, wherein the moving member has conductivity, and wherein the detection object portion comprises a recessed portion provided on the moving member.
7. The position detection device according to claim 1, wherein the plurality of the detection coils comprise a first detection coil and a second detection coil, and wherein the detection range of the first detection coil and the detection range of the second detection coil partially overlap and are continuous in the moving direction.
8. The position detection device according to claim 7, wherein during when the moving member moves from one moving end to an other moving end in the moving direction, each of the first detection coil and the second detection coil faces a plurality of the detection object portions that are provided so as to be spaced apart in the moving direction.

9. The position detection device according to claim 8, wherein the plurality of the detection object portions, which face the first detection coil during when the moving member moves from the one moving end to the other moving end, each have a different length in the moving direction, and wherein the plurality of the detection object portions, which face the second detection coil during when the moving member moves from the one moving end to the other moving end, each have a different length in the moving direction.
10. The position detection device according to claim 1, wherein among the plurality of the detection coils, at least one detection coil and at least one other detection coil have different lengths from each other in the moving direction, and wherein a cycle of an output signal of the at least one detection coil and a cycle of an output signal of the at least one other detection coil during movement of the moving member are different from each other.
11. The position detection device according to claim 10, wherein during when the moving member moves from one moving end to an other moving end in the moving direction, each of the plurality of detection coils faces a plurality of the detection object portions that are provided so as to be spaced apart in the moving direction.